TASK 11: Data Access Layer using Spring Data JPA

AIM

To implement a Data Access Layer using Spring Data JPA by creating a repository with JpaRepository, and performing custom query operations with filtering, sorting, and pagination on student records.

ALGORITHM

1. Start
2. Create Spring Boot project with Spring Data JPA dependency
3. Create Student entity class
4. Create Repository interface extending JpaRepository
5. Define custom query methods (by department, age, etc.)
6. Implement sorting using Sort class
7. Implement pagination using Pageable interface
8. Create service layer to handle logic
9. Test using REST controller / Postman
10. End

Code :

```
<!DOCTYPE html>
<html>
<head>
    <title>Student JPA Dashboard - Final Output</title>
    <style>
        body {
            font-family: Arial;
            background: #f4f4f4;
            text-align: center;
        }
        .container {
```

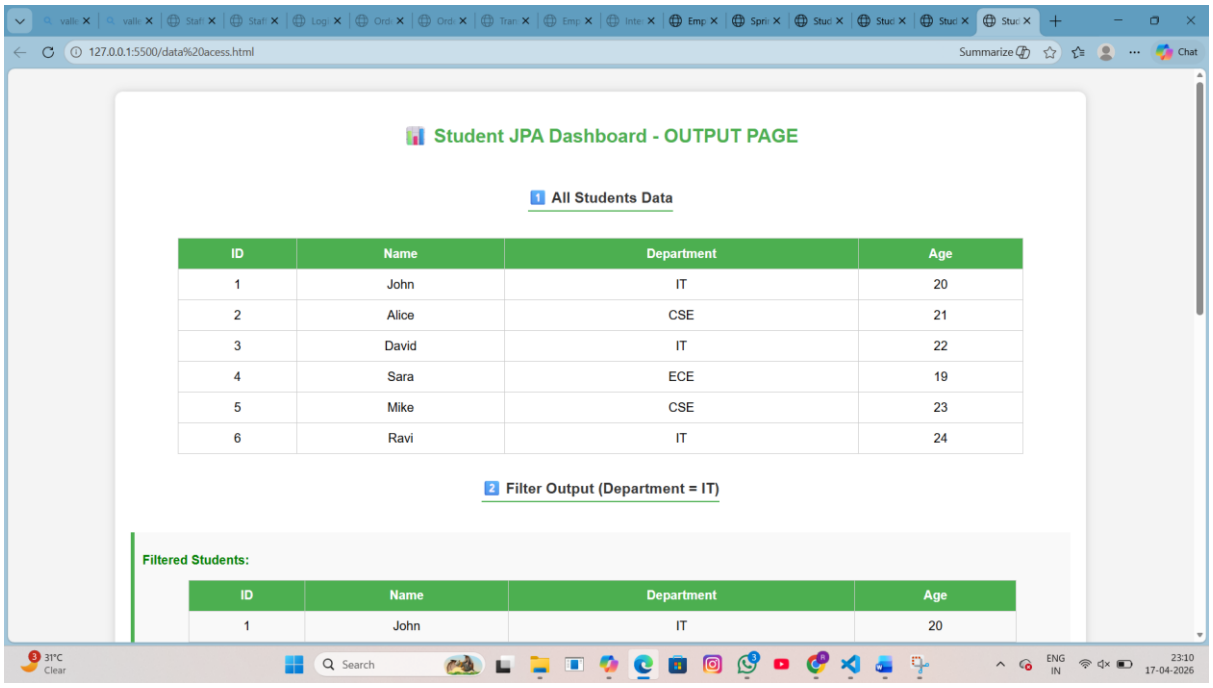
```
    width: 80%;
    margin: auto;
    background: white;
    padding: 20px;
    margin-top: 30px;
    border-radius: 10px;
    box-shadow: 0px 0px 10px #ccc;
}
h2 {
    color: #4CAF50;
}
h3 {
    color: #333;
    margin-top: 30px;
    border-bottom: 2px solid #4CAF50;
    display: inline-block;
    padding-bottom: 5px;
}
table {
    width: 90%;
    margin: auto;
    border-collapse: collapse;
    margin-top: 15px;
}
th, td {
    border: 1px solid #ccc;
    padding: 10px;
    text-align: center;
}
```

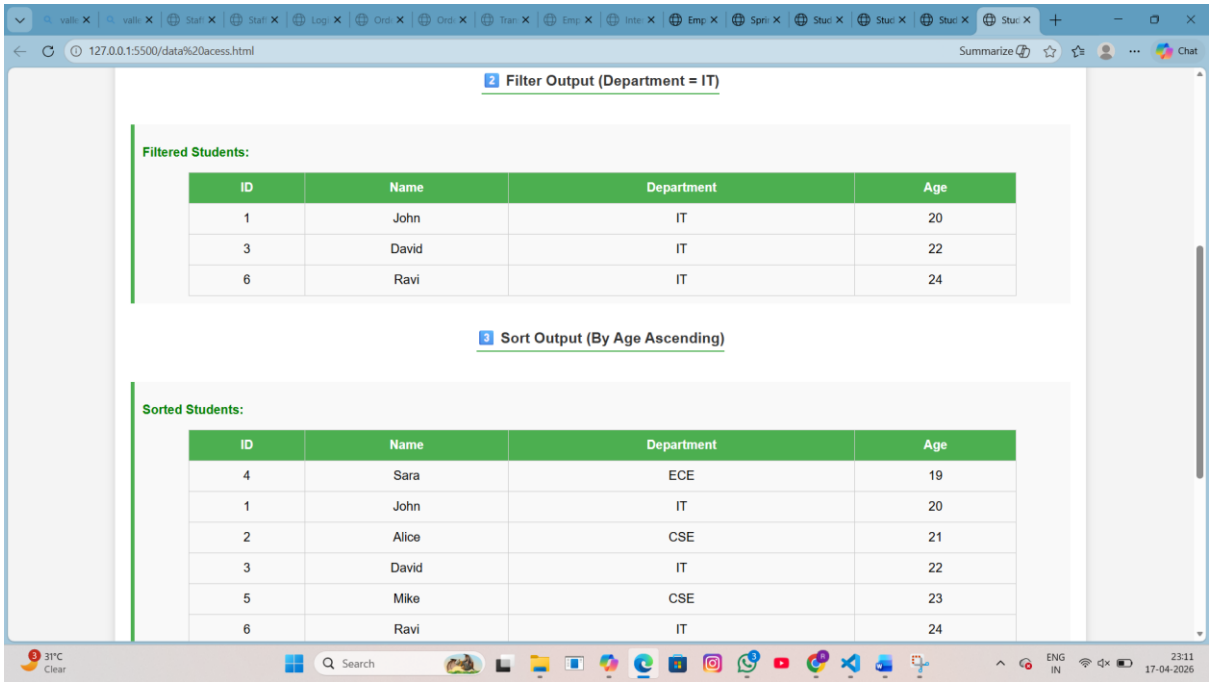
```
        th {
            background: #4CAF50;
            color: white;
        }
        .box {
            margin-top: 20px;
            padding: 10px;
            background: #f9f9f9;
            border-left: 5px solid #4CAF50;
            text-align: left;
        }
        .highlight {
            color: green;
            font-weight: bold;
        }
    </style>
</head>
<body>
<div class="container">
    <h2>📊 Student JPA Dashboard - OUTPUT PAGE</h2>
    <!-- ================= FULL DATA ================= -->
    <h3>1️⃣ All Students Data</h3>
    <table>
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Department</th>
            <th>Age</th>
        </tr>
```

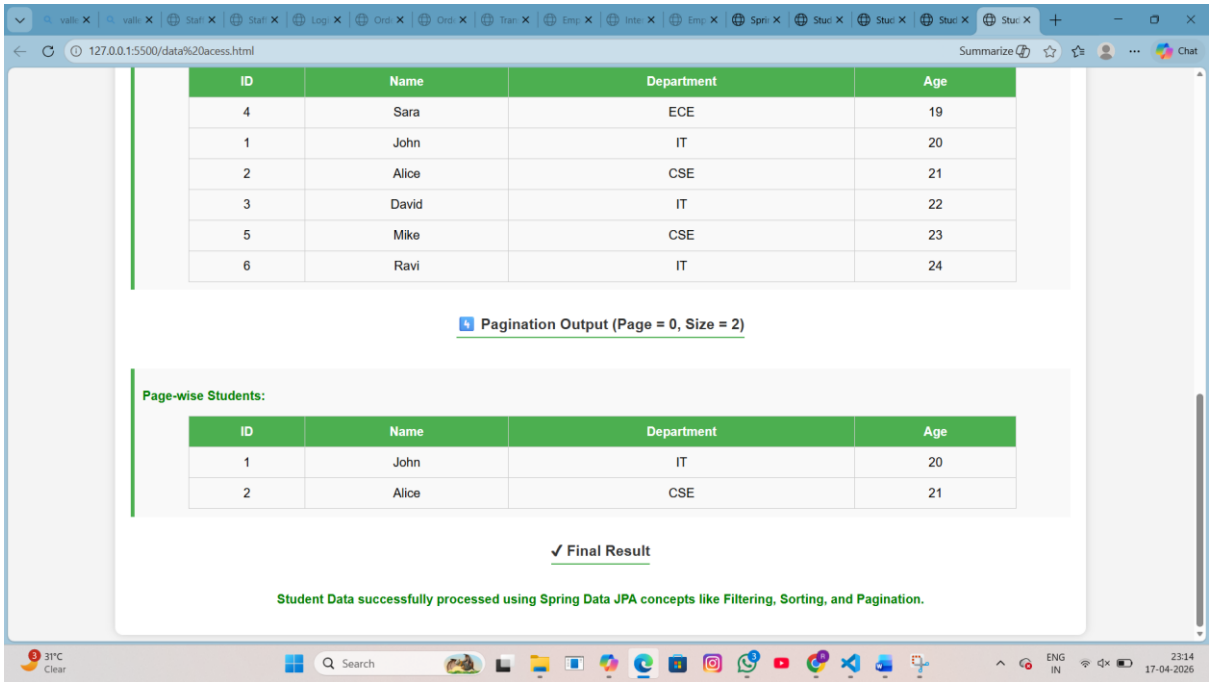
```
    <tr><td>1</td><td>John</td><td>IT</td><td>20</td></tr>
    <tr><td>2</td><td>Alice</td><td>CSE</td><td>21</td></tr>
    <tr><td>3</td><td>David</td><td>IT</td><td>22</td></tr>
    <tr><td>4</td><td>Sara</td><td>ECE</td><td>19</td></tr>
    <tr><td>5</td><td>Mike</td><td>CSE</td><td>23</td></tr>
    <tr><td>6</td><td>Ravi</td><td>IT</td><td>24</td></tr>
</table>
<!-- ================= FILTER OUTPUT ================= -->
<h3>2⃣ Filter Output (Department = IT)</h3>
<div class="box">
    <p class="highlight">Filtered Students:</p>
    <table>
        <tr>
            <th>ID</th><th>Name</th><th>Department</th><th>Age</th>
        </tr>
        <tr><td>1</td><td>John</td><td>IT</td><td>20</td></tr>
        <tr><td>3</td><td>David</td><td>IT</td><td>22</td></tr>
        <tr><td>6</td><td>Ravi</td><td>IT</td><td>24</td></tr>
    </table>
</div>
<!-- ================= SORT OUTPUT ================= -->
<h3>3⃣ Sort Output (By Age Ascending)</h3>
<div class="box">
    <p class="highlight">Sorted Students:</p>
   <table>
        <tr>
            <th>ID</th><th>Name</th><th>Department</th><th>Age</th>
        </tr>
        <tr><td>4</td><td>Sara</td><td>ECE</td><td>19</td></tr>
```

```
            <tr><td>1</td><td>John</td><td>IT</td><td>20</td></tr>
            <tr><td>2</td><td>Alice</td><td>CSE</td><td>21</td></tr>
            <tr><td>3</td><td>David</td><td>IT</td><td>22</td></tr>
            <tr><td>5</td><td>Mike</td><td>CSE</td><td>23</td></tr>
            <tr><td>6</td><td>Ravi</td><td>IT</td><td>24</td></tr>
        </table>
    </div>
    <!-- ================= PAGINATION OUTPUT ================= -->
    <h3>4□Pagination Output (Page = 0, Size = 2)</h3>
    <div class="box">
        <p class="highlight">Page-wise Students:</p>
        <table>
            <tr>
                <th>ID</th><th>Name</th><th>Department</th><th>Age</th>
            </tr>
            <tr><td>1</td><td>John</td><td>IT</td><td>20</td></tr>
            <tr><td>2</td><td>Alice</td><td>CSE</td><td>21</td></tr>
        </table>
    </div>
    <!-- ================= FINAL RESULT ================= -->
    <h3>✓ Final Result</h3>
    <p class="highlight">
        Student Data successfully processed using Spring Data JPA concepts like Filtering, Sorting, and Pagination.
    </p>
</div>
</body>
</html>
```

Output :


127.0.0.1:5500/data%20acess.html
Student JPA Dashboard - OUTPUT PAGE
1 All Students Data
ID | Name | Department | Age
1 | John | IT | 20
2 | Alice | CSE | 21
3 | David | IT | 22
4 | Sara | ECE | 19
5 | Mike | CSE | 23
6 | Ravi | IT | 24
2 Filter Output (Department = IT)
Filtered Students:
ID | Name | Department | Age
1 | John | IT | 20


127.0.0.1:5500/data%20acess.html
2 Filter Output (Department = IT)
Filtered Students:
ID | Name | Department | Age
1 | John | IT | 20
3 | David | IT | 22
6 | Ravi | IT | 24
3 Sort Output (By Age Ascending)
Sorted Students:
ID | Name | Department | Age
4 | Sara | ECE | 19
1 | John | IT | 20
2 | Alice | CSE | 21
3 | David | IT | 22
5 | Mike | CSE | 23
6 | Ravi | IT | 24



RESULT :

A complete Student JPA Dashboard output page was successfully created. It displays all student records along with filtered results, sorted data, and paginated views. This demonstrates how Spring Data JPA operations like filtering, sorting, and pagination are represented in a user-friendly frontend format.